Sreenu Mappompilly Raj

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PROFESSIONAL SUMMARY

Full-stack product engineer with 6+ years building Python backend systems (FastAPI, Django, Flask) end-to-end — from requirement through API design, database modeling, integrations, and production deployment. Strong on SQL/data modeling, async/event-driven processing (Kafka), auth & permissions, and production debugging via Prometheus/Grafana/Splunk. Comfortable with React/TypeScript, enough to own a feature front-to-back.

TECHNICAL SKILLS

Backend: Python (FastAPI, Django, Flask), Go, TypeScript, React/ReactNative

Data & Messaging: PostgreSQL, Redis, DuckDB, Apache Kafka, GraphQL

Cloud & Infra: AWS (EC2, S3, RDS, Lambda, API Gateway, CloudWatch, DynamoDB, SQS, VPC, CloudFront), Docker, Kubernetes, Terraform

CI/CD & Security: GitLab CI/CD, GitHub, Jenkins, SonarQube, OWASP DAST, Trivy

Observability/Testing: Prometheus, Grafana, Splunk · Pytest, Selenium, Playwright, Postman

GenAI (exploratory): Gen AI, LLM, LangChain, LangGraph, Agentic AI, RAG, Claude Code

WORK EXPERIENCE

Ara Solutions Ltd. *Oct 2024 – Present · India*

Software Engineer · Next.js · Prisma · PostgreSQL · Tailwind CSS · Zod · React Hook Form · TanStack Table · better-auth · Vercel

- Built a full-stack Next.js (TypeScript) platform supporting inventory, CRM, invoicing, and service workflows.
- Designed a scalable PostgreSQL + Prisma backend (15+ models) with RBAC, 2FA, validation, and audit logging.
- Engineered production-grade features (server actions, dynamic UIs, integrations) with extensible AI/automation architecture.
- Owned features end-to-end — from requirement through backend/database design, integration, and production deployment.

University of Strathclyde *Jan 2023 – Dec 2024 · Glasgow, UK*

Full-time Postgraduate Study — MSc Advanced Computer Science (Merit)

- Coursework and projects covering distributed systems, cloud architecture, and LLM/AI applications.

Bosch Global Software Technologies *Sep 2019 – Oct 2022 · India*

Software Engineer · Python · FastAPI · Django · Kafka · Docker · AWS · Terraform · GitLab CI · Prometheus

- Designed Python FastAPI microservices with Kafka event-driven communication, improving scalability and decoupling async workloads.
- Built internal Django platform for source code metrics with Grafana dashboards, integrated into Jenkins pipelines.
- Integrated SonarQube (SAST), OWASP (DAST), and Trivy for DevSecOps compliance; instrumented services with Prometheus and Grafana for production monitoring and faster incident debugging.
- Migrated core apps from on-prem to AWS using Terraform — reducing infrastructure costs by 25% and deployment errors by 50%.
- Built full-stack Jira ticket automation tool (React + FastAPI + SSO) to standardise and deduplicate ticket creation.

NOTABLE PROJECTS

Tonic · Go · PWA · Android · Claude Code

- Designed and built a real-time music key/chord detection app end-to-end using Claude Code (AI-assisted development), shipped as a web app, installable PWA, and fully offline native Android app from a single Go codebase.

React Native Mobile Application · React Native · Figma

- Translated Figma design into a production-quality React Native app with reusable components, responsive layouts, and structured navigation.

EDUCATION

MSc Advanced Computer Science · Merit *Jan 2023 – Dec 2024*

University of Strathclyde, Glasgow, UK